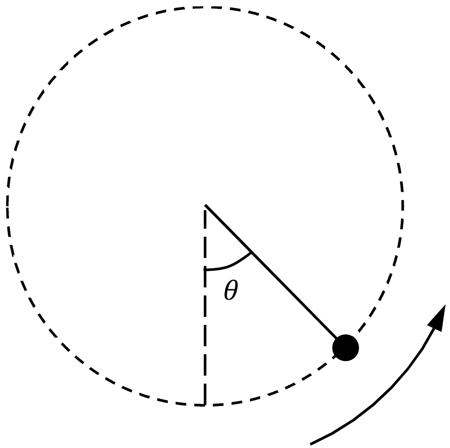
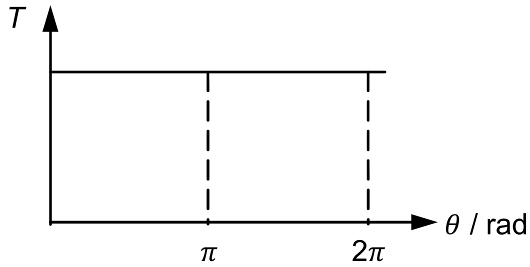
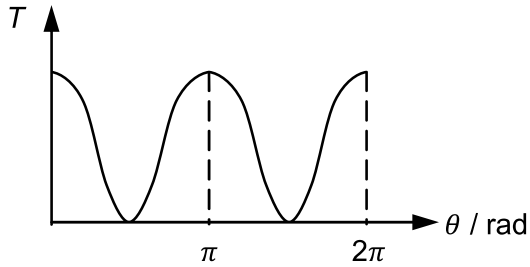
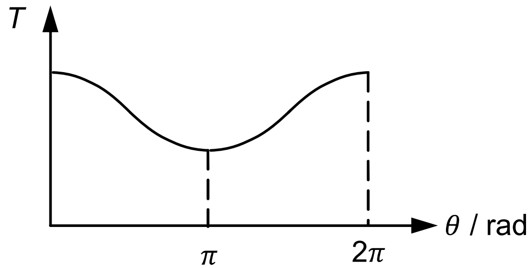
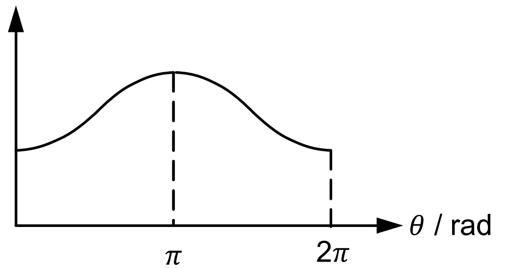
11 An object is fixed to one end of an inextensible light string which rotates in a vertical circle at constant speed.



Which graph could represent the variation with angle θ of the tension T in the string?

[Turn over

	A	B
		
	C	**D**
		

Answer: C

Resultant of force by string and weight of object provides centripetal force

At $\theta = 0$, $\quad T_0 - mg = \frac{mv^2}{r} \Rightarrow T_0 = \frac{mv^2}{r} + mg$

At $\theta = \pi$, $\quad T_\pi + mg = \frac{mv^2}{r} \Rightarrow T_\pi = \frac{mv^2}{r} - mg$

So, tension in string is the least at top of the circle.

12	Two stationary particles of masses M_1 and M_2 are a distance d apart. A third particle, lying